Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

The area above the x -axis enclosed by the curves $x^2 - y^2 = 0$ and $x^2 + y - 2 = 0$ is

(A) $\frac{5}{3}$

(B) $\frac{7}{3}$

(C) $\frac{8}{3}$

(D) $\frac{10}{3}$

Q.2

The area bounded by the x -axis, the curve $y = f(x)$ and the lines $x = 1, x = b$, is equal to $\sqrt{b^2 + 1} - \sqrt{2}$ for all $b > 1$, then $\mathrm{f(x)}$ is

(A) $\sqrt{x - 1}$

(B) $\sqrt{x + 1}$

(C) $\sqrt{x^2 + 1}$

(D) $\frac{x}{\sqrt{1+x^2}}$

Q.3

If the area enclosed by $\mathrm{y^2 = 4ax}$ and line $\mathrm{y = ax}$ is $1/3\mathrm{sq.}$ units, then the area enclosed by $\mathrm{y = 4x}$ with same parabola is

(A) $8\mathrm{sq.}$ units

(B) $4\mathrm{sq.}$ units

(C) $4/3$ sq. units

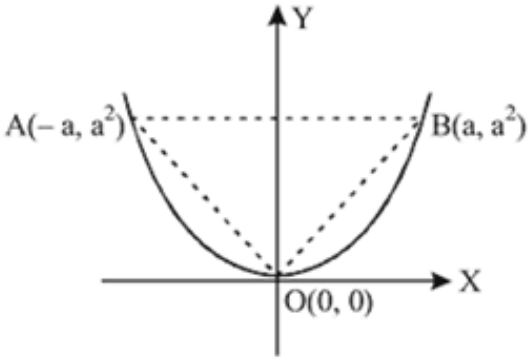
(D) $8/3$ sq. units

Q.4

The figure shows as triangle AOB and the parabola $\mathrm{y = x^2}$. The ratio of the area of the triangle AOB to the

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area of the region AOB of the parabola $y = x^2$ is equal to



(A) $\frac{3}{5}$

(B) $\frac{3}{4}$

(C) $\frac{7}{8}$

(D) $\frac{5}{6}$

Q.5

The area included between the parabolas $y^2 = 4a(x + a)$ and $y^2 = 4b(x - a), b > a > 0$, is

(A) $\frac{4\sqrt{2}}{3}b^2\sqrt{\frac{a}{b-a}}$ sq. units

(B) $\frac{8\sqrt{8}}{3}a^2\sqrt{\frac{b}{b-a}}$ sq. units

(C) $\frac{4\sqrt{2}}{3}a^2\sqrt{\frac{b}{b-a}}$ sq. units

(D) $\frac{8\sqrt{8}}{3}b^2\sqrt{\frac{a}{b-a}}$ sq. units

Q.6

The area common to the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ and $\frac{x^2}{b^2} + \frac{y^2}{a^2} = 1, 0 < b < a$ is

(A) $(a + b)^2 \tan^{-1}\frac{b}{a}$

(B) $(a + b)^2 \tan^{-1}\frac{a}{b}$

(C) $4ab \tan^{-1}\frac{b}{a}$

(D) $4ab \tan^{-1}\frac{a}{b}$

Q.7

The area bounded by $y = x^2 + 3$ and $y = 2x + 3$ is (in sq. units)

(A) $\frac{12}{7}$

(B) $\frac{4}{3}$

(C) $\frac{3}{4}$

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(D) $\frac{8}{3}$

Q.8

The area enclosed between the curves $y = \log_e(x + e), x = \log_e\left(\frac{1}{y}\right)$, and the x -axis is

(A) 2 sq. units

(B) 1 sq. units

(C) 4 sq. units

(D) None of these

Q.9

The area of the figure bounded by $y^2 = 2x + 1$ and $x - y = 1$ is

(A) $\frac{2}{3}$

(B) $\frac{4}{3}$

(C) $\frac{8}{3}$

(D) $\frac{16}{3}$

Q.10

What is the area enclosed between the curves $y^2 = 12x$ and the lines $x = 0$ and $y = 6$?

(A) 2sq unit

(B) 4 sq unit

(C) 6 sq unit

(D) 8 sq unit

Q.11

What is the area bounded by the curves $y = e^x, y = e^{-x}$ and the straight line $x = 1$?

(A) $\left(e + \frac{1}{e}\right)$ sq unit

(B) $\left(e - \frac{1}{e}\right)$ sq unit

(C) $\left(e + \frac{1}{e} - 2\right)$ sq unit

(D) $\left(e - \frac{1}{e} - 2\right)$ sq unit

Q.12

Area bounded by $y = \tan^{-1}x, \quad y = \cot^{-1}x$ and y-axis is equal to

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(A) $\ln \sqrt{2}$ sq. unit

(B) $\ln\ 4$ sq. unit

(C) $\ln\ 8$ sq. unit

(D) $\ln\ 2$ sq. unit

Q.13

The area of the region formed by $x^2+y^2-6x-4y+12 \leq 0, y \leq x$ and $x \leq \frac{5}{2}$ is

(A) $\left(\frac{\pi}{6}-\frac{\sqrt{3}+1}{8}\right)$ sq unit

(B) $\left(\frac{\pi}{6}+\frac{\sqrt{3}-1}{8}\right)$ sq unit

(C) $\left(\frac{\pi}{6}-\frac{\sqrt{3}-1}{8}\right)$ sq unit

(D) None of these

Q.14

Area enclosed between the curves given by $y=x^4$ and $y=x^2+12$ will be (in sq. units)

(A) $\frac{304}{15}$

(B) $\frac{594}{15}$

(C) $\frac{608}{15}$

(D) cannot be determined

Q.15

If $c_1=y=\frac{1}{1+x^2}$ and $c_2=y=\frac{x^2}{2}$ be two curves lying in XY -plane, then

(A) area bounded by curve $y=\frac{1}{1+x^2}$ and $y=0$ is $\frac{\pi}{2}$

(B) area bounded by c_1 and c_2 is $\frac{\pi}{2}-1$

(C) area bounded by c_1 and c_2 is $1-\frac{\pi}{2}$

(D) area bounded by curve $y=\frac{1}{1+x^2}$ and x -axis is $\frac{\pi}{2}$

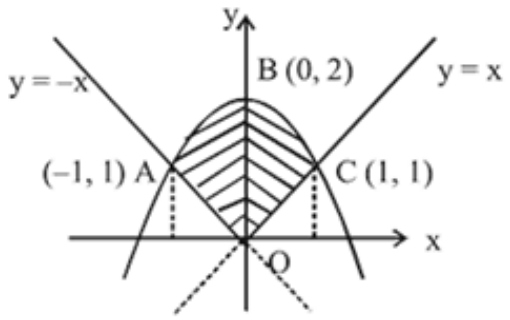
Answers & Solutions

Q.1 Answer:

$\frac{7}{3}$

Solution:

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We first draw the given curves The first curve $x^2 - y^2 = 0 \Rightarrow y = \pm x$ represents a pair of straight lines with slopes 1 and -1 passing through origin. The second curve $\Rightarrow \mathrm{x}^2 + \mathrm{y} - 2 = 0 \Rightarrow \mathrm{x}^2 = -\mathrm{y} + 2 \Rightarrow \mathrm{x}^2 = -(\mathrm{y} - 2)$

represents a parabola with vertex $(0, 2)$ axis as y -axis and concavity dawnwards (see the chapter of parabola in coordinates). Both the curves are plotted in the figure and the required area is shown by the shaded region. The points A and C are the points of intersection of $y^2 = x^2$ with $x^2 + y - 2 = 0$

Solving the two equations, we get $\mathrm{y}^2 + \mathrm{y} - 2 = 0$ [putting value of $\mathrm{x}^2 = \mathrm{y}^2$] $\Rightarrow (y + 2)(y - 1) = 0$

giving $\mathrm{y} = -2$ and 1 , but $\mathrm{y} = -2$ is discarded as the required area is above the x -axis. $\therefore \mathrm{y} = 1 \Rightarrow \mathrm{x} = \pm 1$

The points A and C are respectively $(-1, 1)$ and $(1, 1)$ now due to symmetry Area of the bounded region OABCO

$= 2\times$ Area $\mathrm{OBCO} = 2 \times \int_0^1 \left[\left(2 - \mathrm{x}^2\right) - \mathrm{x}\right] \mathrm{dx}$

[Since $\mathrm{y} = 2 - \mathrm{x}^2$ is the upper curve and $\mathrm{y} = \mathrm{x}$ is the lower curve]

$= 2\left[2\mathrm{x} - \frac{\mathrm{x}^3}{3} - \frac{\mathrm{x}^2}{2}\right]_0^1 = 2\left[2 - \frac{1}{3} - \frac{1}{2}\right] = \frac{7}{3}$

Q.2 Answer:

$\frac{x}{\sqrt{1+x^2}}$

Solution:

Given $\int_1^{\mathrm{b}} \mathrm{f}(\mathrm{x})\mathrm{dx} = \sqrt{\mathrm{b}^2 + 1} - \sqrt{2}$

Differentiate with respect to b

$f(b) = \frac{b}{\sqrt{b^2+1}} \Rightarrow f(x) = \frac{x}{\sqrt{x^2+1}}$

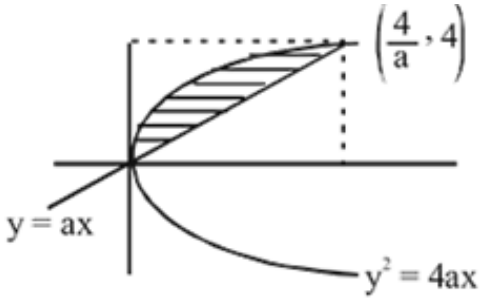
Q.3 Answer:

$8/3$ sq. units

Solution:

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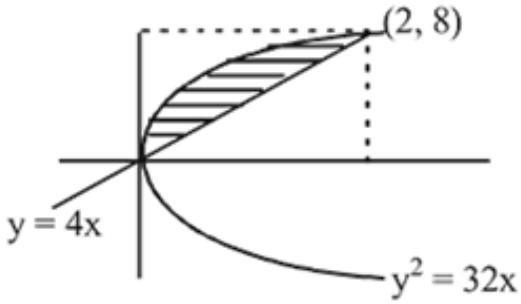
Point of intersection of $y^2 = 4ax$ **and** $y = ax$ **are** $(0, 0)$ **and** $\left(\frac{4}{a}, 4\right)$



Given $\int_0^4 \left[\frac{y}{a} - \frac{y^2}{4a}\right] dy = \frac{1}{3}$

$\Rightarrow \frac{8}{a} - \frac{1}{12a} \times 64 = \frac{1}{3} \Rightarrow \frac{8}{3a} = \frac{1}{3} \Rightarrow a = 8$

So, the parabola is $y^2 = 32x$



Area enclosed by $y = 4x$ **is**

$\int_0^8 \left[\frac{y}{4} - \frac{y^2}{32}\right] dy = \left[\frac{y^2}{8} - \frac{y^3}{96}\right]_0^8 = \frac{8}{3}$

Q.4 Answer:

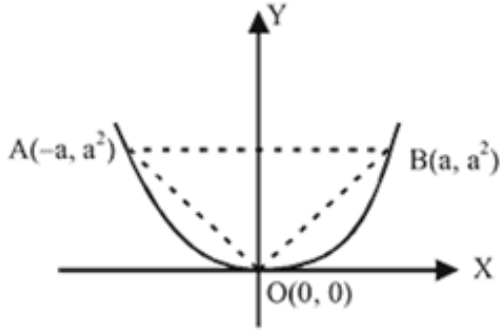
$\frac{3}{4}$

Solution:

Area of $\Delta AOB = \frac{1}{2} \times$ **base** $\times$ **height**

$= \frac{1}{2} \times 2a \times a^2 = a^3$ **units**

Area of region AOB



$= 2\int_0^{a^2} x dy = 2\int_0^{a^2} \sqrt{y} dy$

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$= 2\left[\frac{y^{3/2}}{3/2}\right]_0^{a^2} = \frac{4}{3}a^3$ **units**

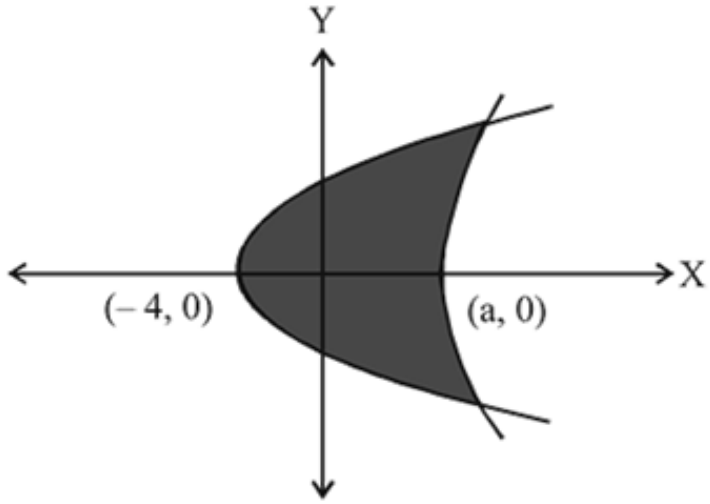
$\therefore$ **ratio of areas** $= \frac{a^3}{\frac{4}{3}a^3} = \frac{3}{4}$

Q.5 Answer:

$\frac{8\sqrt{8}}{3}a^2\sqrt{\frac{b}{b-a}}$ sq. units

Solution:

We have $y^2 = 4a(x+a)\ldots(i)$**, a parabola with vertex** $(-a, 0)$
and $y^2 = 4\,b(x-a)\ldots(ii)$**, a parabola with vertex** $(a, 0)$
Solving (i) and (ii), we get $y = \pm a\sqrt{\frac{8\,b}{b-a}}$



$A = 2\int_0^{\frac{a\sqrt{8b}}{\sqrt{b-a}}}\left(\left(\frac{y^2}{4b}+a\right)-\left(\frac{y^2}{4b}-a\right)\right)dy$

$= 2\int_0^{\frac{a\sqrt{8b}}{\sqrt{b-a}}}\left(2a-\frac{(b-a)y^2}{4ab}\right)dy$

$= 2\left[2ay-\frac{b-a}{12ab}y^3\right]_0^{\frac{a\sqrt{8b}}{\sqrt{b-a}}}$

$= 2\left[2a\times\frac{a\sqrt{8b}}{\sqrt{b-a}}-\frac{b-a}{12ab}\left(\frac{a\sqrt{8b}}{b-a}\right)^3\right]$

$= 4a^2\sqrt{\frac{8b}{b-a}}-\frac{4}{3}a^2\sqrt{\frac{8b}{b-a}} = \frac{8a^2}{3}\sqrt{\frac{8b}{b-a}}$ **sq. units**

Q.6 Answer:

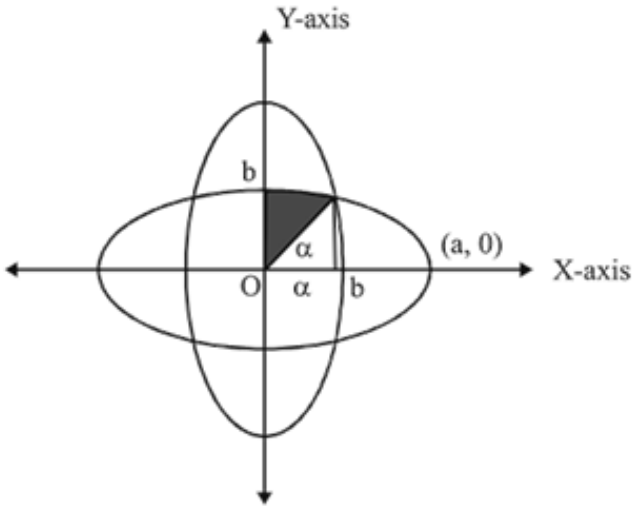
$4ab\tan^{-1}\frac{b}{a}$

Solution:

We have, $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ **and** $\frac{x^2}{b^2}+\frac{y^2}{a^2}=1$ **both are ellipse with centre** $(0,0)$**, vertex** $(a,0),(-a,0)$ **and** $(0,b),(0,-b)$
The curves intersect at $(\pm\alpha,\pm\alpha)$

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where, $\alpha = \frac{ab}{\sqrt{a^2+b^2}}$



The area of shaded region

$\int_0^{\alpha} \frac{b}{a}\sqrt{a^2-x^2}dx - \frac{\alpha^2}{2}$

$= \frac{\mathrm{b}}{2\mathrm{a}}\left[\mathrm{x}\sqrt{\mathrm{a}^2-\mathrm{x}^2} + \mathrm{a}^2\sin^{-1}\frac{\mathrm{x}}{\mathrm{a}}\right]_0^{\alpha} - \frac{\alpha^2}{2}$

$= \frac{\mathrm{b}}{2\mathrm{a}}\left[\alpha\sqrt{\mathrm{a}^2-\alpha^2} + \mathrm{a}^2\sin^{-1}\frac{\alpha}{\mathrm{a}}\right] - \frac{\alpha^2}{2}$

$= \frac{b}{2a}\cdot\frac{ab}{\sqrt{a^2+b^2}}\cdot\frac{a^2}{\sqrt{a^2+b^2}} +$

$\frac{ab}{2}\sin^{-1}\frac{b}{\sqrt{a^2+b^2}} - \frac{a^2b^2}{2(a^2+b^2)}$

$= \frac{ab}{2}\tan^{-1}\frac{b}{a}$

Required area $= 8I = 4\mathrm{ab}\tan^{-1}\frac{\mathrm{b}}{\mathrm{a}}$ **sq. units**

Q.7 Answer:

$\frac{4}{3}$

Solution:

Given curves are $y = x^2 + 3$ **and** $y = 2x + 3$ **points of intersection are** $(0, 3)$ **and** $(2, 7)\therefore$

Required area

$$= \left|\int_0^2 \left(x^2 - 2x\right) dx\right| = \left|\frac{x^3}{3} - \frac{2x^2}{2}\right|_0^2$$

$$= \left|\frac{8}{3} - 4\right| = \frac{4}{3} \text{ sq. unit}$$

Q.8 Answer:

2 sq. units

Solution:

$y = \log_e(x+e), x = \log_e\left(\frac{1}{y}\right)$ **or** $y = e^{-x}$

For $y = \log_e(x+e)$**, shift the graph of** $y = \log_e x, e$

units to the left hand side.

Required area $= \int_{1-e}^{0} \log_e(x+e)dx + \int_0^{\infty} e^{-x}dx$

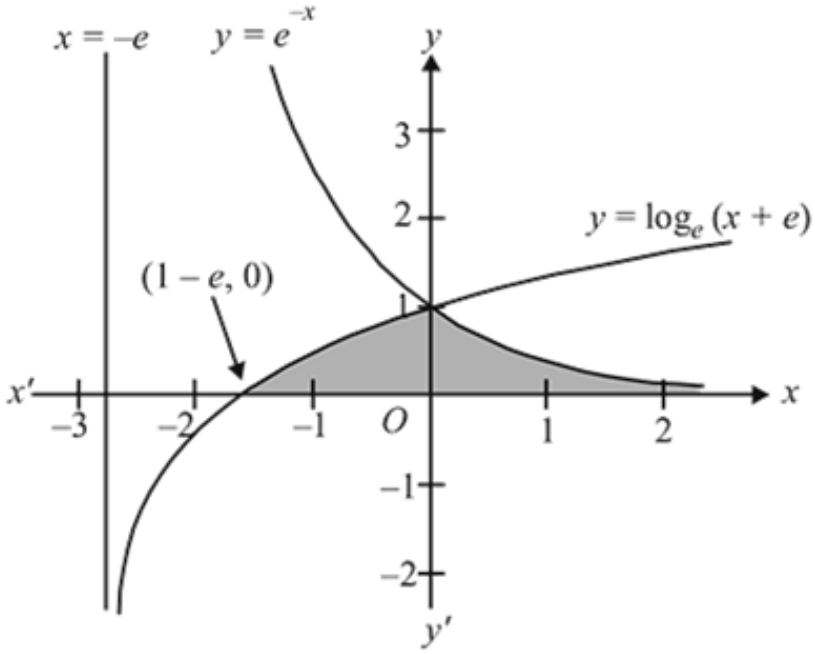
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$= |x \log_e(x+e)|_{1-e}^{0} - \int_{1-e}^{0} \frac{x}{x+e} dx - |e^{-x}|_0^{\infty}$

$= \int_0^{1-e} \left(1 - \frac{e}{x+e}\right) dx - e^{-\infty} + e^0$

$= |x - e\log(x+e)|_0^{1-e} - 0 + 1$

$= 1 - e + e\log e + 1 = 2$ **sq. units.**



Q.9 Answer:

$\frac{16}{3}$

Solution:

To find area bounded by

$y^2 = 2x + 1$ **... (i)**

$x - y - 1 = 0$ **... (ii)**

On solving the equation (i) and (ii),

$x - y = 1$

$y^2 = 2(y-1) + 1$

$y^2 = 2y - 1$

$(y+1)(y-3) = 0$

$y = 3$ **or** -1

$\therefore x = 4, 0$

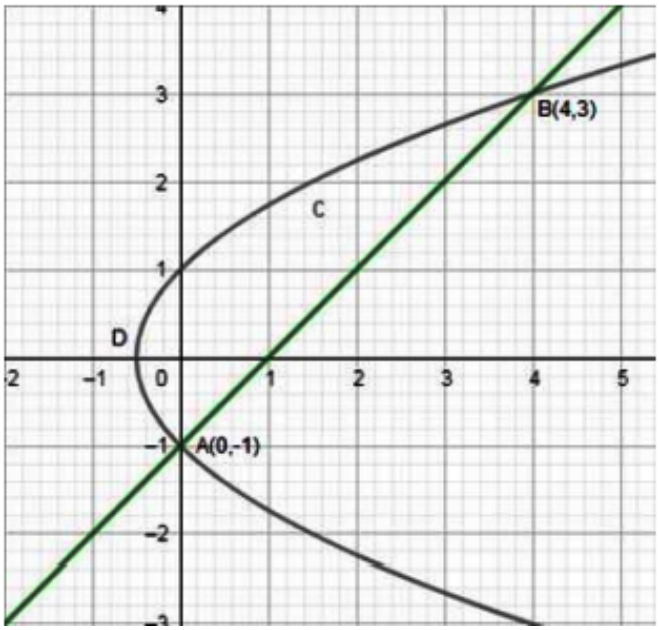
Equation (i) is a parabola with vertex $\left(-\frac{1}{2}, 0\right)$ **and passes through** $(0,1), A(0,-1)$

Equation (ii) is a line passing through $(1,0)$ **and** $(0,-1)$

Points of intersection of parabola and line are $B(4,3)$ **and** $A(0,-1)$

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These are shown In the graph below:



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Required area = Region ABCDA

$= \int_{-1}^{3} \left(1 + y - \frac{y^2-1}{2}\right) dy$

$= \frac{1}{2} \int_{-1}^{3} \left(2 + 2y - y^2 + 1\right) dy$

$= \frac{1}{2} \int_{-1}^{3} \left(3 + 2y - y^2\right) dy$

$= \frac{1}{2} \left[3y + y^2 - \frac{y^3}{3}\right]_{-1}^{3}$

$= \frac{1}{2} \left[(9 + 9 - 9) - \left(-3 + 1 + \frac{1}{3}\right)\right]$

$= \frac{1}{2} \left[9 + \frac{5}{3}\right]$

$= \frac{32}{6}$ **sq. units**

$= \frac{16}{3}$ **sq. units**

Q.10 Answer:

6 sq unit

Solution:

Equation of given curve is $y^2 = 12x$

At $y = 6, 36 = 12x \Rightarrow \quad x = 3$

$\therefore$ **Required area** $= \int_0^3 (y_1 - y_2)\, dx$ **where** y_1 **represents line and** y_2 **represents the curve.** $=$

$\int_0^3 (6 - \sqrt{12x}) dx = [6x]_0^3 - \sqrt{12} \left[\frac{2x^{3/2}}{3}\right]_0^3$

$= [6 \times 3] - \frac{\sqrt{12} \times 2 \times \sqrt{27}}{3} = 18 - 12 = 6$ **sq unit**

Q.11 Answer:

$\left(e + \frac{1}{e} - 2\right)$ sq unit

Solution:

Given equations of curvesare $y = e^x$ **and** $y = e^{-x} \Rightarrow e^x = \frac{1}{e^x} \Rightarrow e^{2x} = e^0 \Rightarrow x = 0$

Also, equation of straight line gives $x = 1$

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$\therefore$ **Required area** $= \int_0^1 (e^x - e^{-x})\,dx$

$= [e^x + e^{-x}]_0^1 = (e + e^{-1}) - (e^0 + e^{-0})$

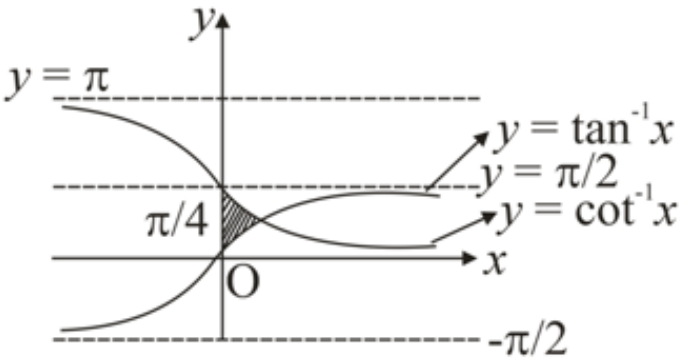
$= \left(e + \frac{1}{e} - 2\right)$ **sq unit**

Q.12 Answer:

ln 2 sq. unit

Solution:

Given curves $y = \tan^{-1}x, \quad y = \cot^{-1}x$.



Required area, $\Delta = \int_0^{\pi/4} \tan y\,dy + \int_{\pi/4}^{\pi/2} \cot y\,dy$

$= \ln \sec y_0^{\pi/4} + \ln \sec y_{5\pi/4}^{\pi/2}$

$= \ln \sec y_0^{\pi/4} + \ln \sec y_{5\pi/4}^{\pi/2}$

$= \ln\sqrt{2} - \ln\frac{1}{\sqrt{2}}$

$= \ln 2$ **sq. unit**

Q.13 Answer:

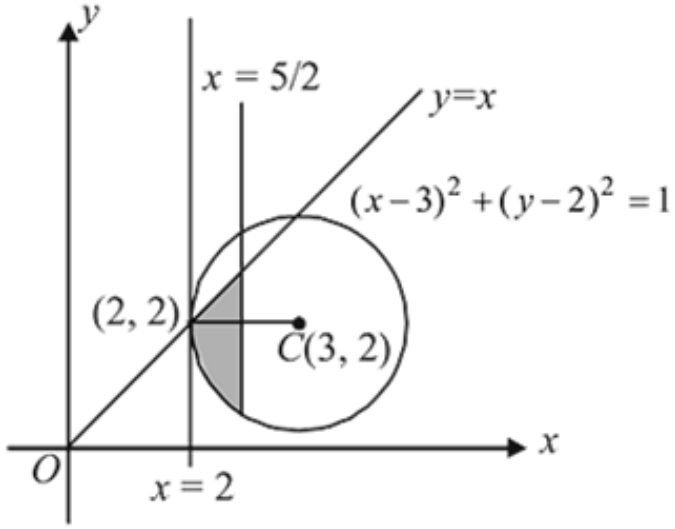
$\left(\frac{\pi}{6} - \frac{\sqrt{3}-1}{8}\right)$ sq unit

Solution:

The required area

$= \int_2^{5/2} x\,dx - \int_2^{5/2} \left[2 - \sqrt{1-(x-3)^2}\right] dx$

$= \left[\frac{x^2}{2}\right]_2^{5/2} - [2x]_2^{5/2} + \left[\frac{x-3}{2}\sqrt{1-(x-3)^2} + \frac{1}{2}\sin^{-1}(x-3)\right]_2^{5/2}$



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$= \frac{9}{8} - 1 + \left(-\frac{\sqrt{3}}{8} + \frac{\pi}{6}\right) = \frac{\pi}{6} - \frac{\sqrt{3}-1}{8}$ **sq unit.**

Q.14 Answer:

$\frac{608}{15}$

Solution:

$f(x) = x^4$

$g(x) = x^2 + 12$

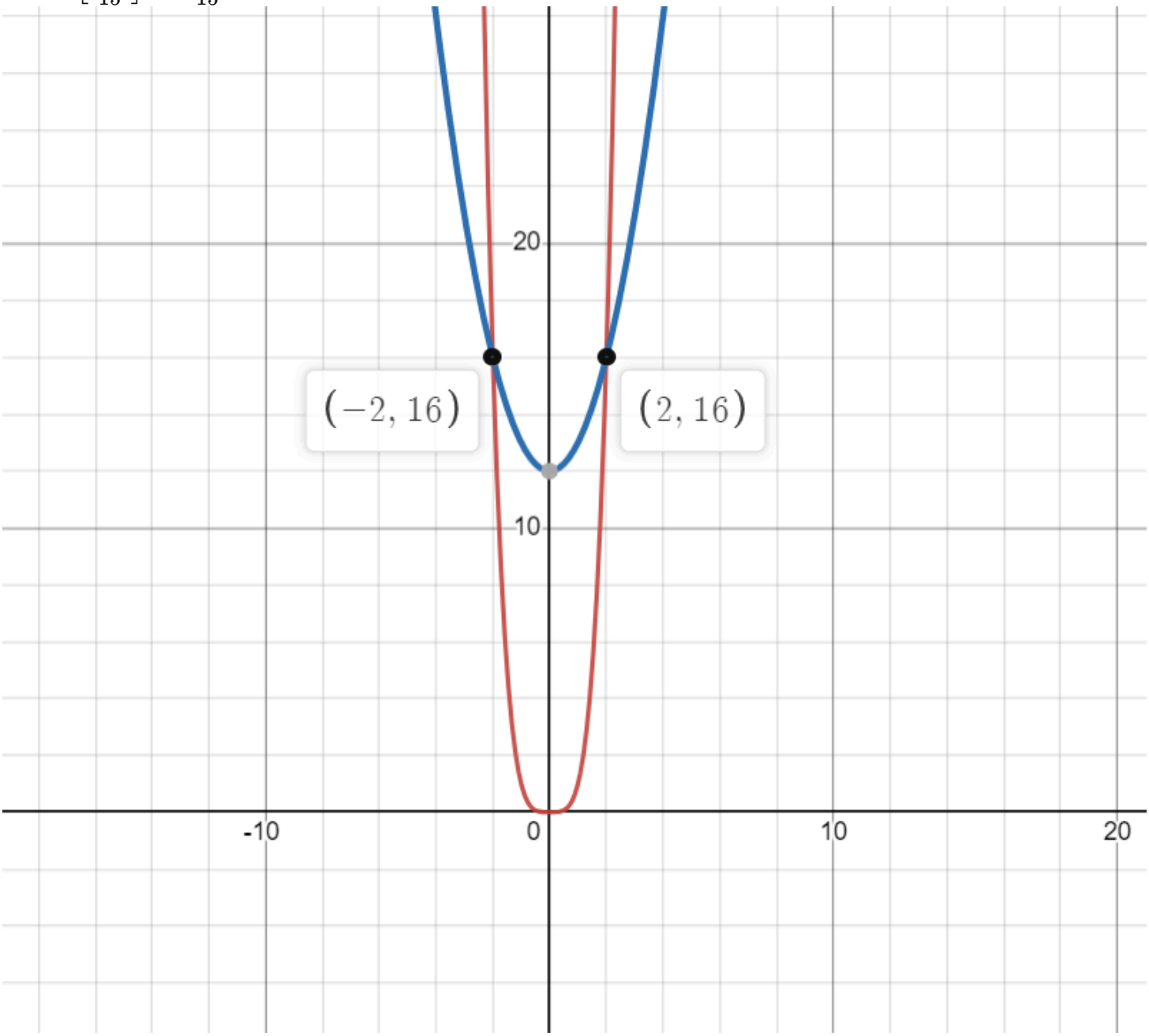
$A = 2 \cdot \int_0^2 [g(x) - f(x)]dx$

$= 2 \cdot \int_0^2 \left(x^2 + 12 - x^4\right) dx$

$= 2 \cdot \left[\frac{x^3}{3} + 12x - \frac{x^5}{5}\right]_0^2$

$= 2 \cdot \left[\frac{8}{3} + 24 - \frac{32}{5}\right]$

$= 2 \cdot \left[\frac{304}{15}\right] = \frac{608}{15}$



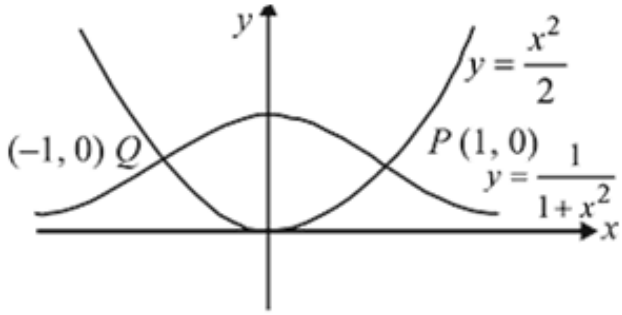
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Q.15 Answer:

area bounded by c_1 and c_2 is $\frac{\pi}{2}-1$

Solution:

Area bounded by $y=\frac{1}{1+x^2}$



and x -axis is $\int_{-\infty}^{\infty}\frac{1}{1+x^2}dx=\pi$

Area bounded by two curves is

$$\int_{-1}^{1}\left(\frac{1}{1+x^2}-\frac{x^2}{2}\right)dx=\frac{\pi}{2}-1$$

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